Grace Brockenbrough, Rose Beall, Annie Jennings - Investments Trading Project

Investment Objectives: Our investor is named Sally and is a 30 year old woman living in DC. She works as a doctor and has a great job. Sadly, her grandfather just passed away and she inherited a million dollars from him. She lived a financially stable life beforehand, but wants her money to grow. Since she had not built the 1 million dollars into her financial plan, this money is a financial cushion not a necessity. She can be a little riskier and invest actively and uniquely. She is about to have kids so growing this money for the long term investments of raising kids could bring her lifestyle to the next level of comfort. She is a relatively younger investor with time on her side and an already financially stable life and job with a higher disposable income, so she has a higher risk tolerance. She would like a portion of this $1M to be safely invested in bonds for short term reliability as she expects to have kids soon, but as for the rest of the money she wants to diversify between reliable value companies and riskier companies with the goal of high growth opportunities for the long term. She sees the world going towards AI and technology more and more every single day both in the real world and in health care so she wants to put a heavy emphasis on this in her equities.

Investment Strategy:

Asset Allocation:

- Equities: **70-80%**
    - Min 50% individual equities (less than 28% in mag 7)
        - Diversify amongst large-cap, mid-cap, small-cap, international
    - 10-15% indexes/ funds
    - 5-10% alternative indexes/funds/ ETFs
- Bonds: 10-20%
- Commodities: 5%

Sally has a high risk tolerance and is young, so her portfolio will be allocated 70–80% in the stock market, 10–20% to bonds, and 5% to commodities. This allocation supports her long-term growth goals while maintaining some balance through lower-risk bond investments and passive index funds. We are considering investments in technology, defense, and health care because Sally is especially interested in these sectors. However, to avoid industry-specific risk, she will diversify across a wide range of industries and global markets. Individual equity investments will be actively managed and focused on strong, long-term companies. Bond investments will be more passive and serve as a stable source of short-term income to help balance the higher risk of equities. We also plan to invest in alternative assets, considering real estate and commodities, to further diversify across asset classes. This provides reliability from assets such as gold and oil, along with the potential growth opportunities of real estate.

Investment Philosophy & Approach: We are going to use a blend of value and growth investing approaches to actively invest in equities in a large portion of our portfolio because Sally has a high risk tolerance for this investment. We will also invest passively into bonds and index funds because that is our stable source of income, less volatile and risky than specific stocks. We are willing to take greater risks for the hope and main goal of growing her wealth by investing in successful and growing companies.

Security Selection:

- Strong and credible management teams
- Positive industry tailwinds
- Long industry runway with sustained growth potential
- Exposure to current events or themes expected to perform well over the next 8 weeks

Other Asset Classes:

- Invest in credible, historically strong index funds
- Allocate capital to bonds for stability
- Consider investments real estate, infrastructure, or natural resources gain exposure to alternatives assets
- Seek opportunities to invest across global markets if we see fit

Trading Strategy:

Buy approach:

- Financial Strength- Companies with consistent growth, strong cash flow, reasonable debt levels, and the ability to invest in their future
- Competitive Edge- Businesses with clear advantages such as brand power, proprietary technology, market leadership, or customer loyalty that help protect long-term profitability
- Business Purpose and Quality- Companies with a clear mission, strong leadership, and alignment with long-term global trends such as technology and healthcare innovation
- Look at 50 and 20 day moving averages to validate initial stock purchase by looking at consistent and growth oriented patterns

<u>Sell approach:</u>

- Major negative business updates
- Shifts in industry direction
- Loss of confidence in leadership or business purpose
- Availability of better opportunities

<u>Risk Management</u>*:* We will diversify across asset classes to protect ourselves from risk. Our portfolio will include both bonds, which are safer, and equities, which are riskier but offer higher growth potential. Within equities, we will also diversify across many different industries to avoid over concentration in one sector and diversify across small vs mid cap. Since we will have individual equity exposure to many of the stocks in the S&P500 we want to lessen our exposure by investing in equal weighted S&P500.

<u>Performance Benchmark</u>*:*

- S&P500, NASDAQ, **Vanguard LifeStrategy Growth Fund (VASGX)**- to represent our portfolio about 80/20 equities vs bonds

<u>Investment Performance Assessment and Reflection</u>

<u>Portfolio's return and Risk:</u> The daily average return on our portfolio was -0.09% and the annualized return was -20.13%. This data says that our investment strategy underperformed and had a significant negative return. The standard deviation of the portfolio was 9.29%, meaning the portfolio returns were about 9.29% away from the mean and there was significant volatility. The Sharpe ratio was about -2.63. For every 1 unit of risk, the portfolio created -2.63 units of excess return (loss) above the risk free rate. This shows that Sally was not rewarded for the risk she took on and would have been better off investing in the risk free. The negative Sharpe ratio does make sense because overall our portfolio lost money, and low or negative Sharpe ratios often highlight bad performance.

It is important when looking at this data to compare and understand it in full context and not just take the numbers for what they are without asking questions. We thought these numbers were very bad and drastic and said our investment strategy was terrible. However, when running the same analysis on our benchmarks such as the S&P, the data was showing a similar story. Over the same period, SPY generated a negative annualized return of -17.62% with higher volatility (13.05%) and a Sharpe ratio of -1.68. This helps give context to such drastic and negative numbers and showcases that the broader market experienced declines and inefficient risk-adjusted performance during the past 8 weeks.

While Sally's portfolio did underperform SPY on raw return (-20.13% vs. -17.62%) and had a more negative Sharpe ratio (-2.63 vs. -1.68), the volatility was significantly lower at 9.29% compared to the market's 13.05%. This is very important because lower volatility means Sally's portfolio was more stable rather than experiencing the sharp swings the broader market saw. For an investor like Sally, who has a higher risk tolerance but also allocated a portion of her portfolio to bonds for stability, this steadiness during a constantly changing market environment is significant. It highlights how a portfolio with diversified asset classes such as bonds and equities helped moderate downside risk. Although the portfolio did not outperform in terms of return, it did a better job of controlling volatility than the market as a whole.

Regression Analysis:

S&P 500: The regression of our portfolio against the SPY tells us important information about how our portfolio moves relative to the SPY. The statistically significant beta of 0.678 means our portfolio is less sensitive to market movements than the SPY, moving 0.678% for every 1% the market moves. This indicates our portfolio carries less systematic risk than the broader market. The alpha on this regression is statistically insignificant, meaning our active management did not generate any excess return above what the SPY already explains. Looking at R-squared, 87.97% of the variance in our portfolio's risk premium is explained by the variance in the SPY's risk premium, reflecting a very strong goodness of fit. This makes sense because most if not almost all of our stocks are components of the SPY. The remaining 12.03% of unexplained variance can be attributed to differences in our allocation and stock selection. The Multiple R of 0.938 confirms that the correlation between our portfolio's risk premium and the SPY's risk premium is very strong.

NASDAQ: The regression against the Nasdaq produces a statistically significant beta of 0.498, meaning our portfolio moves 0.498% for every 1% the Nasdaq moves, reflecting lower sensitivity to the Nasdaq than to the SPY. This makes sense because the Nasdaq is heavily concentrated in technology. While we have significant exposure to the tech industry, our diversification across other industries reduces how closely our portfolio tracks a purely tech-dominated index. The alpha is statistically insignificant with a p-value greater than 0.05, so no excess return is generated beyond what the Nasdaq explains. The R-squared of 0.858 tells us that 85.8% of the variance in our portfolio's risk premium is explained by the variance in the Nasdaq's risk premium, which is a strong goodness of fit but slightly weaker than our fit against the SPY. The Multiple R of 0.926 reflects strong correlation between our portfolio's risk premium and the Nasdaq's risk premium, though again slightly lower than against the SPY, which is consistent because we have diversification in other sectors beyond technology.

Vanguard LifeStrategy Growth Fund (VASGX): This regression is one of the most interesting comparisons because VASGX is the closest to our allocation and investment objectives, investing 80% in stocks and 20% in bonds, which aligns with Sally's target asset allocation. The statistically significant beta of 0.69 is the highest of all three regressions, so our portfolio is most sensitive to movements in VASGX relative to the other benchmarks. For every 1% VASGX moves, our portfolio moves 0.69% with it. This aligns with our assumptions because we chose VASGX as it shares a very similar structure to our portfolio in terms of the balance between equities and fixed income. The alpha is statistically insignificant (p-value greater than 0.05), so our portfolio does not generate excess return beyond what VASGX already captures. The Multiple R of 0.918 is the highest across all three regressions, indicating that the correlation between our portfolio's risk premium and VASGX's risk premium is the strongest of any benchmark we tested. The R-squared of 0.842 means that 84.2% of the variance in our portfolio's risk premium is explained by the variance in VASGX's risk premium, confirming a strong goodness of fit. The largest beta and high R-squared both make sense and reinforce that VASGX is the most comparable benchmark to our portfolio since it is the only benchmark that takes into account the 80/20 ratio of equities and bonds represented in our asset allocation.

Investment Strategy and Asset Allocation

Our buy strategy focused on financial strength, competitive edge, and business purpose and quality, while our security selection focused on strength and credibility of management teams, industry tailwinds and runway, and current events. This approach was strong because it provided incredible flexibility, however there are changes we would make in retrospect.

While we thought through the allocation between different asset classes, we could have been more specific about an allocation breakdown between industry sectors. We aimed to diversify across industries but did not follow very specific guidelines. We often would investigate a company, consider our anticipated future, and invest heavily if we believed it would perform well, which caused us to unintentionally overweight some industries such as technology and some stocks within each industry. Sally, as a high risk tolerance investor looking for growth opportunities in technology-heavy industries, leaned towards a concentrated portfolio, which while this was consistent with her objective, it may not have benefited her overall returns or protection against industry specific risks. In the future, we should have improved our diversification across a greater variety of sectors and company sizes across small, mid, and large capitalization.

Regarding our sell strategy, we built the portfolio explicitly around Sally's long-term objectives. This money serves as extra cushion since she is financially stable, so we allocated bonds to serve as the stable short-term portion of the portfolio in case she needs liquidity, while stocks were designated for the long-term, higher-risk portion. This played out as the stock market was very volatile throughout the year, and we ultimately decided to limit ourselves to little to no selling.

Netflix is a great example. It was under scrutiny regarding a deal with Warner Bros., but as long-term investors, our objective was to try not to sell at the bottom. Netflix ultimately did rebound, which validates our strategy to hold as long term investors. Another example is Chevron, which has been prominent in the news as the political landscape and economy around oil changes daily. Tensions in the Middle East have cut off a huge chunk of the world's oil supply, driving prices way up, and Chevron, as a major American oil company, is taking in massive profits as a result. At various points we considered selling to collect profits, but we remained invested for the long run, and had we sold, we would have missed out on a significant portion of the upside.

We also considered whether we should have introduced more formal selling rules throughout the term. However, we believe this could have worked against us, as rigid rules may have pushed us toward selling in ways that do not align with the overall purpose and needs of our investor. We are glad we did not react to temporary market scrutiny because in retrospect we believe we chose our companies well, and we did not want to act on the irrational sentiments of the market. Within our framework, there is no specific sell strategy other than special and compelling circumstances that would lead us to believe a company is going to experience a significant and sustained decline. We believe that our upfront decision to buy a company is more powerful than any temporary fluctuation or news cycle, as our investor is focused on the long term.

Equity Portfolio Attribution Analysis:

Security Selection

It is important to note that the same equity stocks had higher portfolio returns under equal-weighting, which tells us that our portfolio’s underperformance was mostly due to allocation decisions and not actual stock selection.

Our security selection was primarily qualitative, which provided flexibility but lacked quantitative metrics. The main fault was failing to balance our qualitative research with analysis of reported financial statements, key ratios, and historicals. Incorporating sector-specific ratios would have strengthened our selection process, for growth-oriented companies like SoFi, recurring revenue and operating margin are more meaningful metrics, while for a stable, cash-generating company like McDonald's, profitability ratios like P/E are a better lens. We would have needed to be cognizant that ratios are not consistent across sectors and that using comparable industry-specific measures would have added an insightful quantitative layer to help our qualitative research. That said, our intuitive approach of analyzing company outlook, sector trends, and current news allowed us to identify the intangibles not always expressed in the numbers, which we believe was just as valuable.

To further criticize our security selection process could have leaned too heavily on current news and short-term market conditions, which could work against us as a long-term investor. Because the most volatile and high-growth companies tend to dominate headlines, our news-driven approach naturally pulled us toward trending, high-risk industries rather than a balanced mix of growth and value. For example, we invested in Angiodynamics because it is investing in cancer treatment markets like NanoKnife, and analysts were showing strong individual stock gains and industry gains in the medical sector, but the price has dropped since the beginning of this year[1]. For an investor like Sally with a long time horizon, we should have been more disciplined about not letting current events drive selection too much, and instead spent more time evaluating the valuations of stable value companies alongside our growth picks.

The most significant fault in our security selection was our overconcentration in large-cap stocks and our failure to invest in mid and small-cap companies. While our strategy explicitly called for diversification across market caps, in practice we gave little real exposure to mid and small-cap names. This is a major missed opportunity, because the highest potential for active outperformance lies in exactly those areas. Large-cap stocks are heavily covered and efficiently priced, meaning it is very difficult to find alpha there. Mid and small-cap companies, on the other hand, are less followed by analysts, which is why if we had really focused on looking into this deeper we could have identified investment opportunities that would not show up in an index. This is where active investing can really earn its value, and we could have taken the reins more aggressively instead of defaulting to the safety and familiarity of large caps.

Portfolio Allocation

We allocated using a variety of techniques. We originally started by allocating about $40,000 to individual equities in the Mag 7 using an equal-weight allocation because they are some of the biggest drivers of overall market performance. We then shifted to a more active approach, adjusting position sizes based on perceived risk levels, industry outlook, and our ability to react to news. This resulted in a judgmental, mixed allocation strategy where we did not strictly follow one weighting method.

To evaluate how each approach impacted performance, we calculated returns using equal-weight (EW), market value-weight (MVW), and price-weight (PW) methods. The results were as follows:

- Equal Weight (EW): -4.71%
- Market Value Weight (MVW): -5.99%
- Price Weight (PW): -6.46%
- Our Portfolio: -6.53%

Our allocation strategy focused too much on maintaining overall asset allocation across asset classes rather than optimizing equity allocation, which is reflected in our weaker returns compared to all three methods.

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[1] Reference Appendix-Trade Notes for article

EW performed the best because it spreads risk more evenly, meaning profits and losses are less drastic. Our portfolio had significant exposure to large positions like Microsoft, NVIDIA, Amazon, and Alphabet, which all experienced notable losses, and EW would have mitigated that concentration. For example, AngioDynamics had one of the highest percentage returns at 6.2% but only resulted in about $250 in profit due to its smaller position size, while Apple generated nearly double the profit (about $490) with only a 1.22% return. EW outperformed simply because many of our largest positions were also our worst performers.

MVW was heavily concentrated in large-cap stocks, with Nvidia, Apple, Google, Microsoft, and Amazon accounting for 75.23% of the portfolio. Smaller companies with large percentage losses had only a 1.34% weight, limiting their negative impact, while top performers like Chevron and Netflix contributed 4.48%. MVW partially protected the portfolio from losses in smaller stocks but still suffered due to significant mega-cap exposure, which underperformed with returns ranging from -11% to -16%.

Under the price-weighted method, the largest seven stocks represented approximately 50% of the portfolio. Among these top-weighted stocks, six had negative returns (-13%, -11%, -18.4%, -16.6%, -11%), and the only positive return was minimal at 0.40%. Five of those top seven positions were relatively small investments of approximately $4,000 to $10,000. This showcases a key flaw of price-weighting because it assigns weight based on share price rather than actual capital allocation, so smaller positions disproportionately influence and misrepresent the portfolio's true exposure.

Critical Analysis of Allocation Strategy: Based on the data, equal-weight allocation would have produced the highest return. However, this does not necessarily mean it would have been the best strategy. EW spreads risk evenly across all holdings, reducing concentration and softening exposure to large positions. It also tends to perform better in volatile markets, which was a key characteristic of the market environment during our eight-week trading period. However, our investment strategy was not to choose the most stable allocation method. Sally has a higher risk tolerance and a strong interest in AI and technology, which led us to concentrate more heavily in those sectors. Using a pure equal-weight strategy would have limited our ability to express these views and allocate more capital to higher-conviction investments. In addition, the outperformance of EW is influenced by hindsight bias. We would not have known at the beginning of the investment period that mega-cap technology stocks would underperform. EW performed well in this specific environment largely because it reduced exposure to our largest losing positions. In conclusion, while EW produced better historical results, we chose a judgment-based allocation strategy to maintain flexibility, adjust to market conditions, and align with the investor’s risk profile. We did not want to assign equal weight to all investments, as some carried higher risk and required more selective capital allocation.

Greatest Profit Loss ($): We invested 232 shares at the original price of $234.21 per share in Amazon. Their products include the sale of consumer goods, advertisements, and subscription services with a focus on technology. The last recorded price was $207.24 and a loss of $6,257.88 in our portfolio. Amazon has been under harsh criticism as the company has attracted worrisome news. In January, Amazon was reported to cut 30,000 jobs which started the year off on a hesitant note for investors. The stock declined 12%, which was the worst performance since December 2022 and was ranked the weakest out of the Magnificent 7. Investors are concerned that the company's high CAPEX related to AI will not equal enough returns in the long-run. The $200 billion commitment outlined to be spent on data centers, chips, and other technological advancements combined with a $50 million dollar investment in OpenAI has contributed to an outlook that FCF could turn negative this year while the ROIC has dropped 2.4% in two quarters. However, none of Bloomberg's analysts recommended a sell of the stock, and our group believed in the stability of Amazon as a reputable company that has a strategic long-term orientation in the technology and AI sector. Amazon accumulated to our greatest dollar loss because a part of our strategy was to invest a more significant chunk in the Magnificent 7 companies because of their large market dominance and commitment to AI which is important to Sally[2].

Our Greatest % Loss: SoFi Technology had a 33.36% loss from the original paid price of $25.63 to the ending price of $17.08, which generated a $1,667.25 loss for our portfolio. SoFi is a fintech company that is in the credit service industry that offers a variety of products for members to borrow, save, lend, and invest. Sally wanted to track SoFi in January because of the company's technology exposure, strong leadership, and positive company momentum. Sally also was optimistic about their operating leverage, consistent GAAP profitability, and high forward P/E of 43.01 compared to traditional banks. In January, President Trump proposed a credit card interest rate cap of 10%, and Sally believed that SoFi could benefit as banks will be more selective with lending, and consumers will look for personal loans. She hoped this would offer a premium for a growth investment in a technology-transformative firm. The short selling firm, Muddy Waters confirmed their short selling position in the company and publicly accused SoFi of inaccurate accounting practices, failing to report $312 million of debt, and inflating EBITDA. SoFi threatened legal action against the false accusations which saved it from a price tank. The tank is likely due to a pause of interest rate cuts combined with risk adverse investors who believe the stock is overvalued with a P/E of 58 in March which is vastly higher than the S&P's average of 29. The significant -33.36% loss is due to the stock's relatively low price, so a drop of $8.55 has a significantly more impact than it does for higher-valued stocks. Sally learned that lower-priced stocks come with a greater amount of risk due to their volatility and sensitivity to small percent changes[3].

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[2] https://finance.yahoo.com/news/amazon-shares-drop-12-200-174330905.html
https://finance.yahoo.com/news/200-billion-ai-bet-either-230800195.html

[3] https://finance.yahoo.com/markets/stocks/articles/sofi-technologies-inc-sofi-refutes-183928644.html
https://finance.yahoo.com/news/why-sofi-stock-fell-22-145000872.html

Greatest Profit ($) and % Gain: Sally's greatest profit gain in dollars was Chevron Corp, which she invested in twice in one day. Chevron Corp gained $2,095.08 from her first buy on February 3rd at a price of $177.21 and again she bought more at a price of $177.26, which gained $1,331.25. By the end of the trading period, the stock reached $197.25, which attributed to a 11.59% gain from her February purchase and a %11.56 gain from the second purchase that day. Sally invested in Chevron because while she was interested the most in technology and consumer companies, she knew that she needed to still diversify her portfolio into energy sectors and with the aspiration of growth opportunities. Chevron is seen to be a large cap, profitable, and dividend-paying firm and now has been referred to as a safe oil company to invest in while being ultra sensitive to Middle Eastern events. She saw in the news that events in Venezuela surrounding oil exports rose 15% in January which Sally believed would benefit Chevron's stock price. In even more recent news, oil prices have further climbed as fear over the conflicts in the Persian Gulf and the closure of the Strait of Hormuz compressing oil supply. Sally was very happy that her investment was achieving great returns, so on March 16th she decided to protect her gains while still allowing for upside and placed a trailing stop at $191 since Chevron has now accumulated a relatively large portion of her portfolio. Her investment in Chevron showed that a company intensely tied to the global economy and sensitive to systematic risk, not just firm specific risk, can leverage intense swings in the market to magnify returns[4].

Citation of AI Use:

- Used in the process of researching and understanding companies for the stock selection process
- Used to edit grammar and clarity after our own initial draft
- Used to help brainstorm different investment strategies for Sally
- Research assistant in better understanding companies, indexes, financial terms to assist in our greater understanding while make analysis
- Use AI to criticize and help to improve our invest strategy, our analysis, and report to help get a more holistic and throughout report and understanding

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https://finance.yahoo.com/news/why-sofi-ceo-big-fan-233000215.html

[4]https://www.reuters.com/markets/commodities/venezuela-oil-exports-rise-chevrons-cargoes-more-supply-china-2025-02-04/

https://www.thestreet.com/investing/stocks/chevron-stock-sends-loud-message-as-oil-panic-grips-wall-street